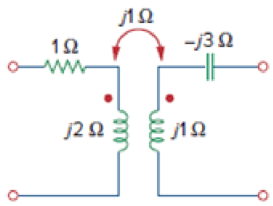
Chapter 19, Problem 51P

Problem

Obtain the *t* parameters for the network in Fig. 19.105.

Figure 19.105



Step-by-step solution

Step 1 of 8

Refer to Figure 19.105 in the textbook for the circuit.

Write the expressions for t-parameters.

$$a = \left.\frac{\mathbf{V}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0} \qquad b = \left.\frac{-\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0}$$

$$c = \left.\frac{\mathbf{I}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0} \qquad d = \left.\frac{-\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0}$$

Here,

$a =$ Open circuit voltage gain.

$b =$ Negative short circuit transfer impedance

$c =$ Open circuit transfer admittance

$d =$ Negative short circuit current gain

Step 2 of 8

To determine a and c, leave the input port open so that $\mathbf{I}_1 = 0$ and place a voltage source $\mathbf{V}_2$ at the output port.

Redraw the circuit as shown in Figure 1.

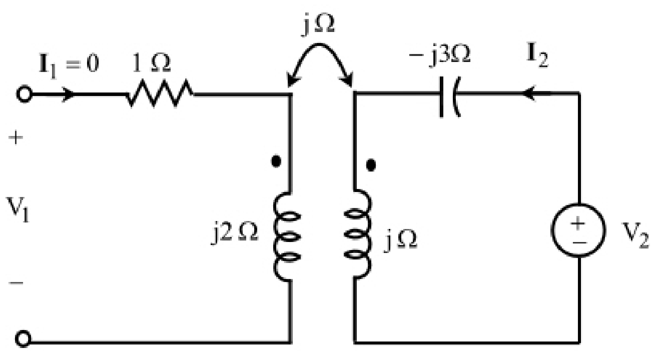


Figure 1

Step 3 of 8

Apply kirchhoff's voltage law at the output port.

$$\begin{aligned}\mathbf{V}_2 &= \mathbf{I}_2(j - j3)\\ &= -j2\mathbf{I}_2\end{aligned}$$

Apply kirchhoff's voltage law at the input port.

$$\begin{aligned}\mathbf{V}_1 &= \mathbf{I}_1(1\,\Omega + j2) + (j\,\Omega)(-\mathbf{I}_2)\\ &= 0 - j\mathbf{I}_2 \qquad\qquad (\text{since } \mathbf{I}_1 = 0)\\ &= -j\mathbf{I}_2\end{aligned}$$

Step 4 of 8

Determine the parameters a and c.

$$\begin{aligned}\mathrm{a} &= \left.\frac{\mathbf{V}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0}\\ &= \frac{-j2\mathbf{I}_2}{-j\mathbf{I}_2}\\ &= 2\end{aligned}$$

$$\begin{aligned}\mathrm{c} &= \left.\frac{\mathbf{I}_2}{\mathbf{V}_1}\right|_{\mathbf{I}_1=0}\\ &= \frac{\mathbf{I}_2}{-j\mathbf{I}_2}\\ &= j\end{aligned}$$

Step 5 of 8

To determine b and d, short circuit the input port so that $\mathbf{V}_1 = 0$ and place a voltage source $\mathbf{V}_2$ at the output port.

Redraw the circuit as shown in Figure 2.

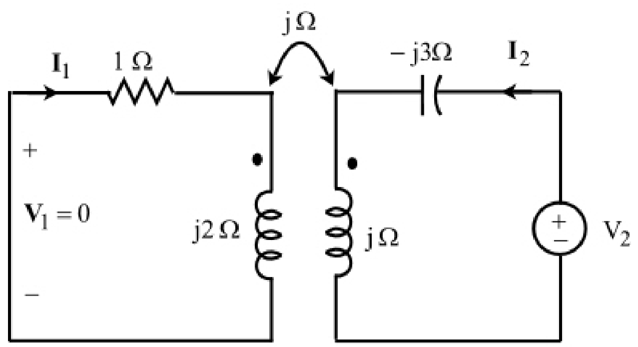


Figure 2

Step 6 of 8

Apply kirchhoff's voltage law to the mesh 1.

$$0 = \mathbf{I}_1(1\,\Omega + j2) - j\mathbf{I}_2$$

$$\mathbf{I}_2 = \mathbf{I}_1\left(\frac{1\,\Omega + j2}{j}\right)$$

$$\frac{\mathbf{I}_2}{\mathbf{I}_1} = 2 - j$$

Step 7 of 8

Apply kirchhoff's voltage law to the mesh 2.

$$\begin{aligned}\mathbf{V}_2 &= \mathbf{I}_2\left(j - j3\right) - j\left(\mathbf{I}_1\right) \\ &= -j2\mathbf{I}_2 - j\mathbf{I}_1\end{aligned}$$

Substitute $(2-j)\mathbf{I}_1$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{V}_2 &= -j2\left(\left(2-j\right)\mathbf{I}_1\right) - j\mathbf{I}_1 \\ &= \mathbf{I}_1\left(-j4-2-j\right)\end{aligned}$$

$$\frac{\mathbf{V}_2}{\mathbf{I}_1} = -j5-2$$

Step 8 of 8

Determine the parameters b and d.

$$\begin{aligned}\mathrm{b} &= \left.\frac{-\mathbf{V}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0} \\ &= -\left(-j5-2\right) \\ &= 2+j5\end{aligned}$$

$$\begin{aligned}\mathrm{d} &= \left.\frac{-\mathbf{I}_2}{\mathbf{I}_1}\right|_{\mathbf{V}_1=0} \\ &= -\left(2-j\right) \\ &= j-2\end{aligned}$$

Therefore, the t-parameters for the network is,

$$\boxed{\begin{bmatrix}\mathrm{a} & \mathrm{b} \\ \mathrm{c} & \mathrm{d}\end{bmatrix} = \begin{bmatrix}2 & 2+5j \\ j & j-2\end{bmatrix}}.$$